

eDNA Data Management in the United States

Reading and References

Suggested Reading

Background on the Science

1. [The ecologist's field guide to sequence-based identification of biodiversity](#) (Creer et al, 2016)
2. [Past, present, and future perspectives of environmental DNA \(eDNA\) metabarcoding: A systematic review in methods, monitoring, and applications of global eDNA](#) (Ruppert et al., 2019)
3. [Critical considerations for communicating environmental DNA science](#) (Stein et al., 2024)
4. [How, What, and Where You Sample Environmental DNA Affects Diversity Estimates and Species Detection](#) (Kirtane et al., 2024)
5. [A Guide to Environmental DNA Extractions for Non-Molecular Trained Biologists, Ecologists, and Conservation Scientists](#) (Rieder et al., 2024)

Data Management and Standards

1. [Publishing DNA-derived data through biodiversity data platforms.](#) (Finstad et al., 2023)
2. [A Practical Approach to Using the Genomic Standards Consortium MIxS Reporting Standard for Comparative Genomics and Metagenomics](#) (Eloe-Fadrosh et al., 2024)
3. [The MIEM guidelines: Minimum information for reporting of environmental metabarcoding data.](#) (Klymus et al. 2024)
4. [Best practices for genetic and genomic data archiving](#) (Leigh et al., 2024)
5. [Centering accessibility, increasing capacity, and fostering innovation in the development of international eDNA standards](#) (Hirsch et al., 2024)

Policy

1. [National Aquatic Environmental DNA Strategy](#) (eDNA Task Team, 2024)

Standards Reference Material

1. [Publishing DNA-derived data through biodiversity data platforms](#). (Finstad et al., 2023)
2. Darwin Core
 1. Quick Reference Guide (term search): <https://dwc.tdwg.org/terms/>
 2. DNA Derived Data Extension: https://rs.gbif.org/extension/gbif/1.0/dna_derived_data_2022-02-23.xml
3. MixS
 1. Intro: <https://www.gensc.org/pages/standards-intro.html>
 2. Term search: https://genomicsstandardsconsortium.github.io/mixs/term_list/

Data Management Tools

1. GBIF Metabarcoding Data Toolkit (MDT): <https://mdt.gbif.org/>
 1. user guide: <https://docs.gbif-uat.org/mdt-user-guide/en/>
2. GBIF-US MDT: <https://mdt.gbif.us/>
 1. manager: Stephen Formel (sformel@usgs.gov)
3. NOAA Omics Data Management Guide: <https://noaa-omics-dmg.readthedocs.io/en/latest/>

Community Pages

1. GBIF: <https://www.gbif.org/dna>
2. OBIS: <https://obis.org/2024/10/22/obis-edna/>

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- eDNA Task Team of the Interagency Working Group on Biodiversity of the Subcommittee on Ocean Science and Technology Committee on Environment of the National Science & Technology Council. 2024. National Aquatic Environmental DNA Strategy. https://www.whitehouse.gov/wp-content/uploads/2024/06/NSTC_National-Aquatic-eDNA-Strategy.pdf
- Eloe-Fadrosch, E.A., Mungall, C.J., Miller, M.A., Smith, M., Patil, S.S., Kelliher, J.M., Johnson, L.Y.D., Rodriguez, F.E., Chain, P.S.G., Hu, B., Thornton, M.B., McCue, L.A., McHardy, A.C., Harris, N.L., Reddy, T.B.K., Mukherjee, S., Hunter, C.I., Walls, R., Schriml, L.M., 2024. A Practical Approach to Using the Genomic Standards Consortium MiX Reporting Standard for Comparative Genomics and Metagenomics, in: Setubal, J.C., Stadler, P.F., Stoye, J. (Eds.), *Comparative Genomics: Methods and Protocols*. Springer US, New York, NY, pp. 587–609. https://doi.org/10.1007/978-1-0716-3838-5_20
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- Kirtane, A., Howard, L., Beaver, C.E., Hunter, M.E., Luikart, G., Deiner, K., 2024. How, What, and Where You Sample Environmental DNA Affects Diversity Estimates and Species Detection. *Environmental DNA* 6, e70042. <https://doi.org/10.1002/edn3.70042>
- Klymus, K.E., Baker, J.D., Abbott, C.L., Brown, R.J., Craine, J.M., Gold, Z., Hunter, M.E., Johnson, M.D., Jones, D.N., Jungbluth, M.J., Jungbluth, S.P., Lor, Y., Maloy, A.P., Merkes, C.M., Noble, R., Patin, N.V., Sepulveda, A.J., Spear, S.F., Steele, J.A., Takahashi, M., Watts, A.W., Theroux, S., 2024. The MIEM guidelines: Minimum information for reporting of environmental metabarcoding data. *Metabarcoding and Metagenomics* 8, e128689. <https://doi.org/10.3897/mbmg.8.128689>

Rieder, J., Jemmi, E., Hunter, M.E., Adrian-Kalchhauser, I., 2024. A Guide to Environmental DNA Extractions for Non-Molecular Trained Biologists, Ecologists, and Conservation Scientists. *Environmental DNA* 6, e70002. <https://doi.org/10.1002/edn3.70002>

Ruppert, K.M., Kline, R.J., Rahman, M.S., 2019. Past, present, and future perspectives of environmental DNA (eDNA) metabarcoding: A systematic review in methods, monitoring, and applications of global eDNA. *Global Ecology and Conservation* 17, e00547. <https://doi.org/10.1016/j.gecco.2019.e00547>

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